

# Signed numbers: addition and subtraction

Initial assessment • Work out all four questions. Show your working.

**1.  $-2 + 9$**

**2.  $6 - 11$**

**3.  $-7 - (-10)$**

**4.  $5 + (-9) - (-6)$**

# Signed numbers: addition and subtraction

Rules recap • Start at the first number.

## Add a positive number

Move right: the value increases.

$$-4 + 7 = 3$$

## Add a negative number

Move left: the value decreases.

$$4 + (-7) = 4 - 7 = -3$$

## Subtract a positive number

Move left: the value decreases.

$$3 - 8 = -5$$

## Subtract a negative number

Move right: subtracting a negative is adding a positive.

$$6 - (-3) = 6 + 3 = 9$$

In  $6 - (-3)$ , the first  $-$  means subtract;  
the  $-$  inside brackets belongs to the number  $-3$ .

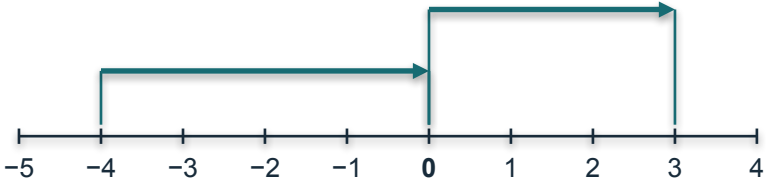
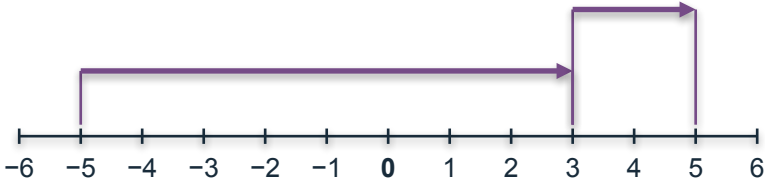
**To subtract a number, add its opposite. Keep the first number unchanged.**

Two negative numbers do not automatically give a positive answer:  
 $-4 + (-3) = -7$ .

# Signed numbers: addition and subtraction

Worked demo • Follow each calculation.

Adding a negative moves left:  $4 + (-7) = 4 - 7 = -3$ .  
Subtracting a negative moves right:  $6 - (-3) = 6 + 3 = 9$ .

|                         | Thread 1: Guided                                                                                                                                                                                                                                                          | Thread 2: Core                                                                                                                                                                               | Thread 3: Depth                                                                                                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The Full Question       | $-4 + 7$                                                                                                                                                                                                                                                                  | $6 - (-3)$                                                                                                                                                                                   | $-5 + 8 - (-2)$                                                                                                                                                                                               |
| Read the Signs          | Start at $-4$ ; add positive $7$ .                                                                                                                                                                                                                                        | Start at $6$ ; subtract negative $3$ .                                                                                                                                                       | Add positive $8$ , then subtract negative $2$ .                                                                                                                                                               |
| Rewrite the Calculation | $-4 + 7$ is already an addition.                                                                                                                                                                                                                                          | $6 - (-3) = 6 + 3$ .                                                                                                                                                                         | $-5 + 8 + 2$ .                                                                                                                                                                                                |
| Calculate               | <div>Move <math>4</math> right to <math>0</math>, then <math>3</math> right: <math>3</math>.</div> <div><math>+4</math> then <math>+3</math>: total <math>+7</math></div> <div></div> | <div><math>6 + 3 = 9</math>.</div> <div>subtract <math>-3 =</math> add <math>3</math></div> <div></div> | <div><math>-5 + 8 = 3</math>; <math>3 + 2 = 5</math>.</div> <div><math>+8</math>, then <math>+2</math></div> <div></div> |
| Check the Answer        | $3 - 7 = -4$ . Answer: $3$ .                                                                                                                                                                                                                                              | $9 + (-3) = 6$ . Answer: $9$ .                                                                                                                                                               | $5 + (-2) = 3$ ; $3 - 8 = -5$ . Answer: $5$ .                                                                                                                                                                 |

# Signed numbers: addition and subtraction

Step-by-step practice 1 • Show your working.

|                         | Thread 1: Guided                                                     | Thread 2: Core                                                       | Thread 3: Depth                                                      |
|-------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|
| The Full Question       | $-6 + 9$                                                             | $4 + (-7)$                                                           | $-8 + 5 - (-4)$                                                      |
| Read the Signs          | Identify the operation signs and the signs belonging to the numbers. | Identify the operation signs and the signs belonging to the numbers. | Identify the operation signs and the signs belonging to the numbers. |
| Rewrite the Calculation | Replace subtraction by addition of the opposite where useful.        | Replace subtraction by addition of the opposite where useful.        | Replace subtraction by addition of the opposite where useful.        |
| Calculate               | Work from left to right. Show each result.                           | Work from left to right. Show each result.                           | Work from left to right. Show each result.                           |
| Check the Answer        | Undo the last operation to check. Write the final answer.            | Undo the last operation to check. Write the final answer.            | Undo the last operation to check. Write the final answer.            |

# Signed numbers: addition and subtraction

Step-by-step practice 2 • Show your working.

|                         | Thread 1: Guided                                                     | Thread 2: Core                                                       | Thread 3: Depth                                                      |
|-------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|
| The Full Question       | $3 - 8$                                                              | $-5 - (-9)$                                                          | $7 - (-2) + (-12)$                                                   |
| Read the Signs          | Identify the operation signs and the signs belonging to the numbers. | Identify the operation signs and the signs belonging to the numbers. | Identify the operation signs and the signs belonging to the numbers. |
| Rewrite the Calculation | Replace subtraction by addition of the opposite where useful.        | Replace subtraction by addition of the opposite where useful.        | Replace subtraction by addition of the opposite where useful.        |
| Calculate               | Work from left to right. Show each result.                           | Work from left to right. Show each result.                           | Work from left to right. Show each result.                           |
| Check the Answer        | Undo the last operation to check. Write the final answer.            | Undo the last operation to check. Write the final answer.            | Undo the last operation to check. Write the final answer.            |

# Signed numbers: addition and subtraction

Independent practice 1 • Choose a column. Show your working and check your answers.

| Thread 1: Guided | Thread 2: Core | Thread 3: Depth      |
|------------------|----------------|----------------------|
| 1. $-3 + 8$      | 1. $5 - (-2)$  | 1. $-4 + 7 - (-3)$   |
| 2. $2 - 7$       | 2. $7 + (-9)$  | 2. $6 - 9 + 2$       |
| 3. $-6 + 4$      | 3. $-4 + (-6)$ | 3. $-5 + (-3) + 10$  |
| 4. $-2 - 5$      | 4. $-8 - (-3)$ | 4. $8 - (-2) - 13$   |
| 5. $-9 + 9$      | 5. $-5 - (-5)$ | 5. $-7 - (-4) + 1$   |
| 6. $4 - 10$      | 6. $0 + (-7)$  | 6. $3 + (-8) - (-5)$ |

# Signed numbers: addition and subtraction

Independent practice 2 • Choose a column. Show your working and check your answers.

| Thread 1: Guided | Thread 2: Core  | Thread 3: Depth       |
|------------------|-----------------|-----------------------|
| 1. $-4 - 6$      | 1. $9 - (-4)$   | 1. $-6 + 9 + (-4)$    |
| 2. $-11 + 11$    | 2. $-6 + (-7)$  | 2. $4 - (-6) + (-10)$ |
| 3. $6 - 15$      | 3. $0 - (-9)$   | 3. $7 - 10 - (-5)$    |
| 4. $-7 + 10$     | 4. $3 + (-8)$   | 4. $-8 - (-2) - 1$    |
| 5. $-8 + 3$      | 5. $-3 - (-8)$  | 5. $-9 + 5 - (-7)$    |
| 6. $5 - 12$      | 6. $-10 - (-6)$ | 6. $5 + (-12) + 3$    |

# Signed numbers: addition and subtraction

Independent practice 3 • Choose a column. Show your working and check your answers.

| Thread 1: Guided | Thread 2: Core  | Thread 3: Depth       |
|------------------|-----------------|-----------------------|
| 1. $-8 + 12$     | 1. $-9 + (-2)$  | 1. $10 - 14 + (-2)$   |
| 2. $-5 - 4$      | 2. $8 - (-5)$   | 2. $6 + (-9) - (-3)$  |
| 3. $-10 + 6$     | 3. $-7 - (-4)$  | 3. $-2 - (-8) + 5$    |
| 4. $9 - 14$      | 4. $6 + (-11)$  | 4. $-3 + (-6) - (-4)$ |
| 5. $7 - 16$      | 5. $-4 - (-10)$ | 5. $9 - (-3) + (-7)$  |
| 6. $-12 + 12$    | 6. $-12 - (-7)$ | 6. $-11 + 8 - 2$      |



# Signed numbers: addition and subtraction

Independent practice 4 • Choose a column. Show your working and check your answers.

| Thread 1: Guided | Thread 2: Core  | Thread 3: Depth       |
|------------------|-----------------|-----------------------|
| 1. $-12 + 7$     | 1. $0 - (-6)$   | 1. $-12 + 7 + 8$      |
| 2. $11 - 18$     | 2. $-11 - (-8)$ | 2. $4 - (-5) + (-12)$ |
| 3. $-6 - 8$      | 3. $5 + (-13)$  | 3. $-6 - (-9) + (-3)$ |
| 4. $-9 + 14$     | 4. $-7 - (-12)$ | 4. $5 - 13 - (-2)$    |
| 5. $-15 + 15$    | 5. $12 - (-3)$  | 5. $8 + (-11) - 4$    |
| 6. $8 - 13$      | 6. $-4 + (-9)$  | 6. $-10 + 6 - (-7)$   |